Characteristic Functions

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Introduction

A **characteristic function** is the Fourier transform of a probability distribution. Unlike the moment generating function, the characteristic function always exists – even for distributions without finite moments, like the Cauchy. Characteristic functions provide the cleanest proofs of classical limit theorems (CLT, LLN) and the most elegant derivations of closure properties of distributional families.

Prerequisites

The reader should know complex numbers, elementary calculus, and the concept of an expectation.

Theory

For a random variable X, the **characteristic function** is

$$\varphi_X(t) = E[e^{itX}] = E[\cos(tX)] + iE[\sin(tX)], \qquad t \in \mathbb{R}.$$

Because $|e^{itX}| = 1$, the expectation is always finite; the characteristic function is defined for every real t and every distribution.

Properties.

- $\varphi_X(0) = 1$ and $|\varphi_X(t)| \leq 1$ for all t.
- $\varphi_{aX+b}(t) = e^{itb}\varphi_X(at)$.
- For independent X, Y: $\varphi_{X+Y}(t) = \varphi_X(t)\varphi_Y(t)$.
- **Uniqueness**: two random variables with the same characteristic function have the same distribution.
- **Inversion**: the density (when it exists) can be recovered from φ by an inverse Fourier transform.
- **Derivatives**: if $E[|X|^k] < \infty$, then $\varphi_X^{(k)}(0) = i^k E[X^k]$.

Continuity theorem (Levy). $X_n \xrightarrow{d} X$ if and only if $\varphi_{X_n}(t) \to \varphi_X(t)$ for every t, provided φ_X is continuous at 0. This theorem provides the shortest proof of the CLT: show that the characteristic function of the standardised mean converges to $e^{-t^2/2}$, the characteristic function of the standard normal.

Examples.

- Normal $\mathcal{N}(\mu, \sigma^2)$: $\varphi(t) = \exp(i\mu t - \sigma^2 t^2/2)$.
- Cauchy: $\varphi(t) = e^{-|t|}$. (The MGF does not exist; the CF does.)
- Poisson(λ): $\varphi(t) = \exp[\lambda(e^{it} - 1)]$.
- Exponential(λ): $\varphi(t) = \lambda/(\lambda - it)$.

Assumptions

Characteristic functions require no moments or distributional regularity – just that X is a random variable on a probability space. This generality is their chief virtue.

R Implementation

```r
set.seed(2026)
n <- 1e5

# Empirical characteristic function of samples from a Cauchy
x <- rcauchy(n)
t_vals <- seq(-3, 3, length.out = 61)
cf_empirical <- sapply(t_vals,
                       function(t) mean(exp(1i * t * x)))

cf_theoretical <- exp(-abs(t_vals))

data.frame(t           = t_vals,
           theoretical = cf_theoretical,
           Re_empir    = Re(cf_empirical),
           Im_empir    = Im(cf_empirical))[seq(1, 61, by = 10), ]
```

We estimate the characteristic function of the Cauchy distribution from a sample and compare to the known form $e^{-|t|}$.

Output & Results

```
       t  theoretical  Re_empir   Im_empir
1   -3.0    0.0498     0.0512     -0.0006
11  -2.0    0.1353     0.1358     -0.0017
21  -1.0    0.3679     0.3689      0.0023
31   0.0    1.0000     1.0000      0.0000
41   1.0    0.3679     0.3689     -0.0023
51   2.0    0.1353     0.1358      0.0017
61   3.0    0.0498     0.0512      0.0006
```

The empirical CF closely matches the theoretical $e^{-|t|}$; the imaginary part is near zero because the Cauchy distribution is symmetric.

Interpretation

Characteristic functions are rarely used directly by applied practitioners but appear everywhere in theoretical proofs. They are the canonical tool for proving that a sequence of distributions converges to a specific limit.

Practical Tips

- When the MGF fails (heavy tails), reach for the characteristic function.
- For computing convolutions (sums of independent variables), multiplying CFs is often simpler than convolving densities.
- Empirical characteristic functions estimated from data can be used for parameter estimation in specific classes of models (stable distributions, alpha-stable processes).
- Inversion to recover densities can be done numerically via FFT; this is how many specialised distributions (e.g., alpha-stable) are computed in R.
- The continuity theorem (Levy) is the standard tool for proving convergence in distribution; many textbook CLT proofs rely on it.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Scientific process and research workflow
- R, RStudio, Quarto, and renv toolchain
- Data types, tidy data, accuracy and precision
- Import, joins, and missingness with dplyr